CS405 - Database Programming using Oracle 11g

Final Term short Notes
Made by Muhammad Ayaz

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Simple Loop

The simple loop is different from other loops available in other programming languages; simple loop consists of a structure to repeat statements
Without exit condition considers to be an infinite loop.
Required to exit the loop

```
DECLARE
rep number(10):=0;
BEGIN
LOOP
if rep >=5 then
dbms_output.put_line ('Value of Rep: ' || rep);
EXIT;
END IF;
rep:=rep+1;
END  LOOP;
END;
```

While Loop

First check the condition and then other loop we be executed

```
DECLARE
```

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```
counter number (4) := 1;
BEGIN
WHILE counter <= 8 loop
dbms_output.put_line (counter);
IF counter=4 then
Exit; -- Loop will be terminated
END IF;
counter := counter +1;
End Loop;
END
```

Do-While Loop

A DO-While loop is a statement that executes a statement at least once and then repeatedly executes the block, or not, depending on a given condition at the end of the block.

We not write the Do keyword in this just we write the loop like simple Loop

Not required to exit the will automatically end the loop if the condition is false

Implementing Do While Loop

```
DECLARE
current_val number (10):=0;
BEGIN
LOOP
dbms_output.put_line ('Value of count is : ' || current_val);
current_val:=current_val+1;
exit when current_val> 10 ;
END LOOP;
END;
```

Numeric FOR Loop

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A FOR LOOP is a repetition control structure that allows you to efficiently write a loop that needs to execute a specific number of times. It's called Numeric because it requires an integer as its terminating point.

Starting limit to upper limit

Not jump in the value in the for loop

If the first value of the for loop is less then the upper value the control move to the for loop body

```
DECLARE
i NUMBER:= 100;
BEGIN
FOR i IN 1..10 LOOP
dbms_output.put_line ( ' Value of I : ' ||i);
i:=i+1; --Error
END LOOP;
end;
```

FOR Loop with Reverse Option

By default, iteration proceeds from the initial value to the final value, generally upward from the lower bound to the higher bound. You can reverse this order by using the REVERSE keyword.

```
DECLARE
loop_start Integer: = 1;
BEGIN
FOR i IN REVERSE loop_start..5
LOOP
DBMS_OUTPUT.PUT_LINE ('Loop counter is ' || i);
END LOOP;
```

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```
END;
```

Continue When Statement

Continue-When statement is also use to unconditionally exit from current iteration of the loop. As in the case of continuous statement, statements after continues-when statement is also skipped and the control is transferred to the next iteration.

```
DECLARE
x NUMBER := 0;
BEGIN
LOOP
DBMS_OUTPUT.PUT_LINE ('Inside loop: x = ' || TO_CHAR(x));
x := x + 1;
CONTINUE WHEN x < 3;
DBMS_OUTPUT.PUT_LINE
('Inside loop, after CONTINUE: x = ' || TO_CHAR(x));
EXIT WHEN x = 5;
END LOOP;
DBMS_OUTPUT.PUT_LINE (' After loop: x = ' || TO_CHAR(x));
END
```

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Nested Loops Loop

can be nested with any other or same type of loops. A nested loop is a loop within a loop, an inner loop within the body of an outer one. How this works is that the first pass of the outer loop triggers the inner

loop, which executes to completion. Then the second pass of the outer loop triggers the inner loop again.

The nested loop or inner loop run completely in a time

```
BEGIN
FOR v_outerloopcounter IN 1..2
LOOP
FOR v_innerloopcounter IN 1..4
LOOP
DBMS_OUTPUT.PUT_LINE('Outer Loop counter is ' || v_outerloopcounter);
DBMS_OUTPUT.PUT_LINE(' Inner Loop counter is ' || v_innerloopcounter);
END LOOP;
END LOOP;
END;
```

Cursor In SQL

Cursor is a Temporary memory or Temporary work Station

A SQL Cursor is a database object used to retrieve the Data from a Reset set one Row at a time

Types of Cursor

Implicit cursor

This Cursor is Created Automatically by Oracle Server Whenever SQL Statement are executed And the user is unware

Explicit Cursor

Are defined by a programmer for going more control over the context Area and Cursor Attribute

Step To create a cursor

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1 Declare the cursor

2 open the Cursor
3 Fetch
4 Close the Cursor
5 De allocating cursor

Declare Cursor_name Scroll For

Select *from Table_name

Method of Cursor

Next for the Next Row

Prior For the Previous Row
First Fort the First Row
Last For the Last Row
Absolute to Find the Specific Record

Relative n

For example

Fetch First From Cursor_name

Fetch Prior From Cursor_Name

Fetch last From Cursor_Name

Attribute

% Found

Is Return true if Next Row is Available in Cursors to process

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%NOT Found

The opposite of Found Next value check if Find false if not found the return True

% Isopen

A cursor is not when is Already open Check open or not check Statements

% RowCount

Return numeric value count How Many Record well change the count of the DML command return only last action

How many row well affected in the last DML command they find the count of the Row

```
DECLARE
cursor c1 is select hiredate, deptno,job from emp where job like ('%MAN%');
doh emp.hiredate%type;
dno emp.deptno%type;
jd emp.job%type;
BEGIN
open c1;
LOOP
fetch c1 into doh,dno,jd;
dbms_output.put_line('Date of hiring: ' || doh );
dbms_output.put_line('Department no: ' || dno );
dbms_output.put_line('Job: ' || jd );
EXIT WHEN c1%notfound;
END LOOP;
END;
```

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Errors

Errors can be defined as compile time syntax issues

Exceptions

exception is a PL/SQL error that is raised during program

Need for Exception Handling

Exception cause the program to terminate the program abnormally and that is not practically possible to foresee all such problematic events.

How Exceptions Handlin Works

Whenever a run-time error is generated, an exception is raised. The flow or the normal execution of the program stops. Control is transferred to the EXCEPTION block of the PL/SQL where the necessary actions are defined to handle exception.

Types of Exceptions

Internal Exceptions: These are pre-defined and they are executed in case of violations of any database rule by the program. Internal exceptions are raised automatically.

User-Defined: PL/SQL also allows user to define their own exceptions according to the need of the program. This type of exceptions is not raised automatically and user should raise it.

Below is the basic syntax of an exception:

Declare Section

All the declarations

Begin

Executable statements

Exception Section – One per block

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Exception handling statements

End

Example

```
Declare
name varchar2 (30):='';
Begin
SELECT ename into name
FROM emp
WHERE empno=9882;
dbms_output.put_line(Name);
Exception
WHEN no_data_found then
dbms_output.put_line('Record not matched');
WHEN others then
dbms_output.put_line(SQLCODE);
dbms_output.put_line(SQLERRM);
End;
```

User Defined Exception?

Any exception defined by the user in order to avoid any specific problematic situation or event are known as User Defined Exceptions. User must have to define them explicitly as they are not available by default in the system.

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The syntax of user defined exception is as follows:

DECLARE

user_define_exception_name EXCEPTION;

BEGIN

statement(s);

IF condition THEN

RAISE user_define_exception_name;

END IF;

EXCEPTION

WHEN user_define_exception_name THEN

User defined statement (action) will be taken;

END;

Write a PL/SQL block to insert row in emp table, while inserting data in emp table if salary is

less than minimum salary then raise an exception other.

Solution:

DECLARE

low_sal EXCEPTION;

min_sal NUMBER:= 10000;

new_sal NUMBER:= 8000;

BEGIN

INSERT INTO EMP(EMPNO, DEPTNO, SAL)

VALUES (4000,20,new_sal);

IF new_sal < min_sal THEN

RAISE low_sal;

END IF;

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EXCEPTION

WHEN low_sal THEN

DBMS_OUTPUT.PUT_LINE ('Salary is less than '||min_sal);

WHEN OTHERS THEN

DBMS_OUTPUT.PUT_LINE (SQLERRM);

END

Raise_Application_Error

It's a built-in procedure lets you issue user-defined ORA- error messages from stored subprograms. That way, you can report errors to your application and avoid returning unhandled exceptions.

Raise vs Raise_Application_Error

Raise is used to call pre-defined or user-defined exception while on the other hand Raise_application_erorr let the developer show the customized message with number. PL/SQL Block terminates the processing when there is some error. If we want to raise an exception and change the path of processing developer can place RAISE statements. By Raise statement, developer can raise user defined exceptions

Declare

name emp.ename%type;

salary emp.sal%type;

Begin

select ename, sal into name, salary from emp where empno=7369;

if salary <5000 then

Raise_application_error(-20030, 'Invalid salary');

else

dbms_output.put_line('Valid Salary');

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End if;

Exception

when others then

dbms_output.put_line(SQLERRM);

End;

. What is exception_init Pragma

The EXCEPTION_INIT pragma associates a user-defined exception name with an error code. Ora – Error no can be intercepted and specific handler can be written. In other words, it allows you to handle the Oracle predefined message by your own message which means you can instruct compiler to associate the specific message to oracle predefined message at compile time.

DECLARE

user_define_exception_name EXCEPTION;

PRAGMA EXCEPTION_INIT(user_define_exception_name,-error_number)

What is Collection?

A collection is an ordered group of elements, all of the same type. It is a general concept that encompasses lists, arrays, and other familiar data types. Each element has a unique subscript that determines its position in the collection and data is accessed through index which can be random. Although collection can hold multiple data but still size of collection is dynamic

Associative arrays

Associative arrays are sets of key-value pairs, where each key is unique and is used to locate a corresponding value in the array. The key can be an integer or a string. Assigning a value using a key for the first time adds that key to the associative array

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Records

Records is a composite datatype which means that it can hold more than piece of imformation to save it a time .

Types of Record

Table Base

Cursor Base

User Defined

Table Based

Declare

Emp-rec emp% rowtype;

Begin

Select * into emp_rec From emp

Where emno = 89;

Dbms_output.put_line(emp_rec.Name || emp_rec.job);

Exception when no-data-found Then

Dbms_output.put_line('noting');

End;

What is Sub-Program

Subprogram created to perform a particular task. These subprograms are combined to form larger programs

What is Procedure?

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Procedure is a database object and a type of subprogram which is created to perform a certain task.

Procedure can performs one or more tasks

Procedure may or may not return value

Procedures are normally used for executing business logic.

```
CREATE OR REPLACE PROCEDURE name_proc
AS
cursor c1 is select * from emp where ename like ('%AK%');
c2 c1%rowtype;
BEGIN
loop
fetch c1 into c2;
dbms_output.put_line (c2.ename);
exit when c1%notfound;
end loop;
END name_proc;
```

Output of the Program:

Procedure Created

/

Calling the Procedure:

```
Begin
Name_proc;
End;
```

two parameters procedure

IN Parameter

An IN parameter lets you pass a value to the Procedure.

It is a read-only parameter.

Inside the subprogram, an IN parameter acts like a constant. It cannot be

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assigned a value

Out Parameter

It returns value to the calling program.

Inside the subprogram, an OUT parameter acts like a variable. You can change its value and reference the value after assigning it.

Function

Function is a database object and a type of subprogram which is created to perform a certain task. A function is a named PL/SQL Block which is similar to a procedure

- It can receive zero or more parameters as an input.
- It can return value to the calling environment.
- Returning value is Mandatory. In other words, Procedure may or may not return value
- whereas function should return one value
- Function are normally used for computation

Function vs. Procedure

Function vs. Procedure comparison, there are following points:

- Both are Database Objects.
- Both can receive one or more parameters
- Procedure can performs one or more tasks whereas function performs a specific task.
- Procedure may or may not return value whereas function should return one value.
- Functions are normally used for computation whereas procedures are normally used for
- executing business logic

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```
CREATE [OR REPLACE] FUNCTION function_name [ (parameter
[,parameter]) ]
RETURN return_datatype
BEGIN
< function_body >
Return value
END function_name;
```

Trigger

A trigger is a database object that automatically executes a specified action or set of actions in response to certain database events, such as inserting, updating, or deleting data from a table

```
CREATE TRIGGER Trigger_name

AFTER INSERT

ON Table_Name

FOR  EACH  ROW

BEGIN

   INSERT INTO Backup_Table (:old.ID, :old.Name, :old.Salary);

END;
```

1. **Row Level Triggers**:
    - Fired for each row affected by the triggering statement.
    - For instance, in an UPDATE statement that modifies multiple rows, a row-level trigger is invoked once for each row modified.

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- If the triggering statement doesn't affect any rows, the row-level trigger is not executed for that statement.

2. **Statement Level Triggers**:
   - Fired once on behalf of the triggering statement, regardless of the number of rows affected.
   - For example, if a DELETE statement removes several rows from a table, a statement-level DELETE trigger is triggered only once, regardless of the number of rows deleted

Dropping the Triggers

The syntax is as follows:

drop trigger Trigger_Name;

What is package?

It is Database Object which group together different Database Objects. groups logically related PL/SQL types, variables, constants, subprograms, cursors, and exceptions. A package is compiled and stored in the database.

Step-1:

CREATE PACKAGE Name of Package AS

Prototype of Database Object (Function, Procedure, Trigger)

End Package Name

Step-2:

CREATE PACKAGE Body Name AS

Implemtation of Prototype of Database Object (Function, Procedure, Trigger) defined in

Package Specification

End Database Object Name

End Package Name

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